

# EVAN DASTIN-VAN RIJN

## Data Scientist and Engineer

📞 401-787-8534   @ theonlydvr@gmail.com   🌐 evandvr.me  
📍 Minneapolis, MN

## EXPERIENCE

PhD Research 11/2021 - Present

Translational Neuroengineering Lab Minneapolis, MN

Working on a thesis titled "Black-box optimization of cognitive control with neuromodulation"

- Developed a real-time Bayesian optimization platform to rapidly and robustly determine ideal stimulation parameters for enhancing highly-noisy cognitive variables
- Developed Bayesian hierarchical models to decode latent cognitive effects of electrical stimulation on behavior
- Developed OSCAR, a hardware interface for communicating with commercial operant chambers from open-source software
- Developed pybehave, a streamlined software platform for integrating behavioral tasks with electrophysiological recordings, video, and digital/analog hardware

Support Data Scientist 05/2024 - Present

Randolph Lab Minneapolis, MN

I'm currently working part time supporting the Randolph Lab in their efforts to analyze a large database of medical records, childhood experiences, and psychological assessments for adopted children. My responsibilities include data visualization, statistical analysis, and the creation of machine learning models to assist in interpreting the database.

AI Training 05/2024 - Present

Outlier and DataAnnotation Remote

I do part-time work training STEM LLMs through Outlier and DataAnnotation in various areas including programming, statistics, and physics. I've completed hundreds of tasks through both services including prompt creation, evaluating model responses, and reviewing other workers' tasks.

Research Assistant 09/2018 - 08/2021

Borton Laboratory Providence, RI

Assisted with a study working towards the development of an adaptive system to treat symptoms of OCD using deep brain stimulation DBS

- Developed Period-based Artifact Reconstruction and Removal Method (PARRM), a novel approach to neurostimulation artifact removal showing improved performance over existing, state-of-the-art methods
- Developed Period Estimation of Lost Packets (PELP), a solution to temporal alignment of packet losses occurring during recordings with implanted bidirectional stimulators
- Developed a neural and video data preprocessing pipeline for accurate temporal alignment of multimodal data streams and data-quality assurance

Various Teaching Experience 08/2018 - Present

- I've worked as a teaching assistant for 5 courses with hundreds of students
- I've helped with creating problem sets, fielding questions, and guiding students through open ended design projects
- I've assisted with lessons in high-school mathematics courses in low-income communities
- I've mentored adults with learning disabilities on topics of their choosing

## SKILLS

### Programming Languages

MATLAB · Python · Java · HTML/CSS · JavaScript · Git · Arduino (C++)

### Analysis Techniques

Bayesian Optimization · Gaussian Process Regression · Hierarchical Modeling · MCMC ·

Clustering · Generalized Linear Mixed Effects Modeling · Vector Graphics

### Laboratory Techniques

Electrophysiology · PCB design · Rat Surgery · Rat Handling · Operant Conditioning

## EDUCATION

### PhD in Biomedical Engineering

University of Minnesota

08/2021 - Present

### Sc. B. in Biomedical Engineering and Neuroscience

Brown University

08/2017 - 05/2021

## STRENGTHS

### 💡 Self-starter

I'm able to quickly identify and pick up new skills and techniques for my work

### 👤 Flexible Work Style

I've lived in 5 countries and have become very comfortable adapting to new work environments and cultures

## KEY ACHIEVEMENTS

### ✍️ Authored >10 peer-reviewed research articles

Substantial experience communicating and visualizing scientific data

### 💰 Secured over \$200,000 in individual research funding

I'm able to effectively communicate the value of my work to funding bodies

### 🔗 Track record of developing scientific software

I'm able to effectively develop and document useful software for aiding scientific research